

How much could AI efficiency change the future if we cut token waste in half?

Posted by u/N-Innov8 • 0 points • 16 comments

Current AI models burn through massive computational cycles repeating context and re-processing redundant tokens, an invisible layer of waste that adds up across billions of interactions.

Global AI data centers already spend over \$450 billion a year and consume 400+ TWh of electricity, projected to double by 2030.

I've been exploring a system-level approach to reduce this token redundancy, potentially making AI conversation engines 50% more efficient.

If this kind of optimization were scaled globally, how do you see it reshaping the future of AI infrastructure, sustainability, and economics?

Curious how the Futurology community envisions the impact of truly efficient intelligence.

Comments

u/[deleted] • 1 points • 2025-10-12 20:28 UTC

[removed]

u/somethingtc • 1 points • 2025-10-12 12:23 UTC

you kick the can down the road for a couple more months. nothing more. the issue GenAI is facing is one of diminishing returns in terms of cost for performance. a "50% increase in efficiency" (which is a number you have pulled from nowhere but i will run with) may sound like a lot but doesn't translate to a doubling of performance.

If you'd like a practical example of this, look at gpt-5 - this latest model was not a performance upgrade, in fact the results are worse in most respects. it was an efficiency upgrade because they are desperate to keep "innovating" but are hitting increasingly high costs.

To put it simply, you're pulling the lever on a slot machine that has a payout of \$100 and costs \$10 for the first spin but the cost doubles every time. you get 5 spins before you can no longer afford to keep going. Now lets say you manage to halve the cost of the spins, a massive improvement in efficiency! surely we can improve our spin number! and absolutely we can, but only to 6. the "future" of GenAI as it stands has not improved markedly at all even if these efficiency numbers were real

u/Poly_and_RA • 1 points • 2025-10-12 11:54 UTC

My thinking is that a 50% change in token-efficiency is near meaningless and would make very close to no difference at all in how AI is used.

It's still worth doing of course, summed up halving the power-usage of AI would be a huge win. But for most use-cases halving resource-use of running queries makes no real difference at all to anything.

u/N-Innov8 • 1 points • 2025-10-12 13:01 UTC

Thanks, consuming less energy resources and reduced impact on the planet would or wouldn't make a difference?

u/Poly_and_RA • 2 points • 2025-10-12 13:36 UTC

I mean any change is a change, but the field of AI is in such huge flux that it's really hard to say where we'll be on anything when the dust settles, progress slow down and we're in a new more stable situation.

At one extreme it's possible that we've now picked most of the low-hanging fruit and that further progress in AI will be slow and modest so that for example most tasks AI can't do today it can also not do 20 years from now.

At the other extreme it's also possible that AI will become good enough that we can use an AI to write a better AI, rinse and repeat until we arrive at a full-blown ASI-powered singularity in which case *ALL* bets are very thoroughly off.

Or we might end up somewhere in between. I consider that to be the most likely outcome.

The closer we get to the second scenario here though, the less things like exactly how many TWhs current AIs are using will matter.

u/wrd83 • 2 points • 2025-10-12 10:39 UTC

I honestly don't care about token efficiency.

Unless AI can generate a high quality result it doesn't matter if I get 5 or 100 tokens in charges.

u/Prior-Enthusiasm2497 • 1 points • 2025-10-12 10:27 UTC

I'm not even sure what the use case is for AI. Shut all those data centers down and what have we lost?

u/N-Innov8 • -3 points • 2025-10-12 10:10 UTC

This post looks at how breakthroughs in AI efficiency, specifically reducing redundant computation at the token level, could alter the long-term future of global infrastructure.

If AI systems can achieve the same performance with half the energy and compute, the implications for sustainability, data-center design, and climate strategy could be profound.

I'd like to discuss how such efficiency shifts might influence the broader future of intelligent systems, economically, environmentally, and socially.

u/twack3r • 5 points • 2025-10-12 11:53 UTC

This post looks at nothing. You propose a 50% increase in efficiency that you are 'working on' (sure thing lil bud lol).

The results would be a 50% increase in efficiency. This would be used to process twice the amount of tokens within the same time span. End of story.

u/N-Innov8 • 0 points • 2025-10-12 13:03 UTC

Halving the cost, halving the global AI energy consumption and reduced carbon footprint? This is just the beginning, not an end in itself!

u/beekersavant • 2 points • 2025-10-12 20:27 UTC

I would think that efficiency as token waste is one aspect. The other is using models with lower parameter count that are more accurate. The models are getting trained, but a UI to take a larger model (10 billion or more parameters) and ask questions, ID used parameters and cut to that for task specific deployments does not really exist. Those things are bespoke and require engineers atm. However, this will be an overriding efficiency. Even, limited deepseek can be run locally for basic programming, but I cannot take that and trim it myself easily. That will be the "revolution" in efficiency.

A small company running an LLM at an interactive register, then easily telling a UI to id all parameters used and cut from 15 billion to as small as possible to create easy local applications. It is the UI that needs to be made. Initial energy output is not as important as being to cut sustained usage to the minimum.

u/N-Innov8 • 1 points • 2025-10-12 20:50 UTC

We have achieved something novel and revolutionary in this context. We shall announce it as early as this week.

u/beekersavant • 2 points • 2025-10-12 21:16 UTC

Who are you? Saying that is not really convincing in the current climate but the opposite. 90% of the engineers I have spoken with regarding creating usable applications do not understand barrier to adoption properly.

u/N-Innov8 • 1 points • 2025-10-12 21:36 UTC

That's a fair question. I'm a research scientist, inventor and founder working on distributed AI systems.

My current work involves reducing redundant computation in conversational AI, optimizing how models manage state and context, rather than retracing entire histories.

I'll share technical details once it's public, but I'm curious how others here see efficiency evolving in multi-model or memory-persistent AI.

u/beekersavant
• 1 points •
2025-10-12
22:06 UTC

Send me a dm when you release. I might have some followups.

As you can see, I have been thinking about the gulf between consumers and model creation. Optimization is the key for local models aimed at the public instead of businesses (even services that run them for the public or say for certain professionals not in the software.) mosy of my field is not going to be able to run a local model - even a relatively small one.

